

1. Administrative & Study Identification

Full Study Title: A Phase II, Randomized Study of Atezolizumab (Anti-PD-L1 Antibody) and Trastuzumab in Combination With Capecitabine and Oxaliplatin (Xelox) in Patients With HER2 Positive Locally Advanced Resectable Gastric Cancer of Adenocarcinoma of Gastroesophageal Junction (GEJ) **Short Title/ Acronym:** ML42058 / Atezolizumab + Trastuzumab + XELOX in GC **Protocol Number:** ML42058 **NCT Number:** NCT04661150 **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Sponsor Name:** F. Hoffmann-La Roche Ltd / Genentech, Inc. (Roche) **Investigational Product:** Atezolizumab (Tecentriq), Trastuzumab (Herceptin), Capecitabine, Oxaliplatin (XELOX) **Phase of Development:** Phase II (PHASE2) **Medical Monitor:** Clinical Operations Lead, F. Hoffmann-La Roche Ltd

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of adding atezolizumab to trastuzumab plus XELOX as measured by the pathological complete response (pCR) rate (defined as no evidence of vital residual tumor cells on hematoxylin and eosin evaluation of the complete resected gastric/GEJ specimen and all sampled regional lymph nodes) following completion of neoadjuvant systemic therapy (NAST) and surgery. **Secondary Objectives:** To evaluate the Objective Response Rate (ORR) during neoadjuvant systemic therapy (NAST), Major Pathological Response (MPR), R0 resection rate, Event-Free Survival (EFS), Disease-Free Survival (DFS), Overall Survival (OS), and overall safety/tolerability (Incidence of Adverse Events).

2.2 Background & Rationale To address the lack of a standard perioperative regimen for locally advanced ERBB2 (HER2)-positive gastric cancer (GC) and gastroesophageal junction (GEJ) cancer. While trastuzumab combined with capecitabine and oxaliplatin (XELOX) is beneficial in HER2-positive GC, and the addition of PD-1/PD-L1 blockade has shown promise in unresectable/metastatic settings (e.g., KEYNOTE-811), its efficacy as a perioperative (neoadjuvant and adjuvant) treatment to improve pCR rates in potentially resectable disease remains to be established.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Trastuzumab + XELOX) **Blinding:** Open-label

Randomization: 1:1 (Arm A: Atezolizumab + Trastuzumab + XELOX vs. Arm B: Trastuzumab + XELOX)

3.2 Planned Enrollment & Duration Target **\$N\$:** 42 participants
Number of Sites: 8 study sites in China **Estimated Study Start:** 25-Feb-2021 **Estimated Primary Completion:** Oct-2023 (Clinical Cutoff: 12-Mar-2023)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 to ≤ 75 years; Chinese ethnicity. 2. Histologically confirmed diagnosis of locally advanced, resectable gastric cancer or adenocarcinoma of the GEJ (Siewert I-III). 3. HER2-positive status defined as either an immunohistochemistry (IHC) score of 3+ or IHC 2+ with amplification proven by in situ hybridization (ISH ≥ 2.0) as assessed by local review based on pretreatment endoscopic biopsies. 4. Clinical stage at presentation: cT3/T4a/T4b, or N+, M0 as determined by AJCC staging system, 8th edition. 5. Availability of pretreatment tumor specimen for biomarker analysis by central lab. 6. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 7. Life expectancy ≥ 12 weeks. 8. Adequate hematologic and end-organ function. 9. For female patients of childbearing potential, agreement (by patient) to remain abstinent (refrain from heterosexual intercourse) or to use highly effective form(s) of contraception during the treatment period and to continue its use for at least: i) 5 months after the last dose of atezolizumab, ii) 7 months after the last dose of trastuzumab, or iii) 6 months after the last dose of capecitabine or oxaliplatin, whichever is longer. 10. For men: agreement to remain abstinent (refrain from heterosexual intercourse) or use a condom, and agreement to refrain from donating sperm.

4.2 Exclusion Criteria 1. Stage IV (metastatic) or unresectable gastric/GEJ cancer determined by investigators. 2. Prior systemic therapy for treatment of gastric cancer. 3. History of malignancy other than GC within 5 years prior to screening, with the exception of malignancies with a negligible risk of metastasis or death. 4. Cardiopulmonary dysfunction. 5. Dyspnea at rest. 6. Active or history of autoimmune disease or immune deficiency. Exceptions: (a) autoimmune-mediated hypothyroidism on thyroid-replacement hormone; (b) controlled Type 1 diabetes mellitus on insulin; (c) eczema, psoriasis, lichen simplex chronicus, or vitiligo with dermatologic manifestations only (rash covering $< 10\%$ BSA, well controlled at baseline requiring only low-potency topical corticosteroids, and no acute exacerbations requiring strong systemic therapies or radiation in the past 12 months). 7. History of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, idiopathic pneumonitis, or evidence of active pneumonitis on screening chest CT scan. 8. Active

tuberculosis. 9. Patients with active hepatitis B. 10. Patients with active hepatitis C.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * Arm A:

Atezolizumab 1200 mg IV (Day 1) + Trastuzumab 6 mg/kg IV (Day 1; 8 mg/kg loading dose) + Oxaliplatin 130 mg/m² IV (Day 1) + Capecitabine 1000 mg/m² orally twice daily (Days 1-14).

Repeated every 3 weeks. * Arm B: Trastuzumab + Oxaliplatin + Capecitabine at identical doses and schedule. **Route of**

Administration: Intravenous (Atezolizumab, Trastuzumab, Oxaliplatin) and Oral (Capecitabine). **Duration of Treatment:** 3 neoadjuvant cycles (3 weeks each) prior to surgery, followed by 5 adjuvant cycles.

5.2 Concomitant & Prohibited Therapy Permitted: Standard pre-medications for chemotherapy-induced nausea and vomiting.

Prohibited: Concurrent investigational therapies, other systemic anti-cancer treatments, or live vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy (HER2 Testing), Baseline Scans
Neoadjuvant Phase	Cycles 1-3	Day 1 of each 21-day cycle: Dosing, Safety Labs. Days 1-14: Oral Capecitabine
Surgery	Post-Cycle 3	Surgical Resection, Pathological Assessment for pCR/MPR
Adjuvant Phase	Cycles 4-8	Day 1 of each 21-day cycle: Dosing, Safety Labs. Days 1-14: Oral Capecitabine
Follow-up	Q12W	Survival tracking, Disease recurrence monitoring (Median follow-up 15.7 months)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pathological Complete Regression (pCR) Rate, defined as no evidence of vital residual tumor cells on hematoxylin and eosin evaluation of the complete resected gastric/GEJ specimen and all sampled regional lymph nodes following the completion of neoadjuvant systemic therapy (NAST) and surgery. **Measurement Method:** Central/Local Pathological Evaluation of the resected surgical specimen.

7.2 Secondary & Exploratory Endpoints * Event-Free Survival (EFS). * Disease-Free Survival (DFS). * Overall Survival (OS). * Major Pathological Response (MPR) rate. * Objective Response Rate (ORR) during neoadjuvant systemic therapy. * R0 resection rate.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of Treatment-Emergent Adverse Events (TEAEs) assessed by CTCAE criteria (e.g., monitoring for grade ≥ 3 TEAEs such as weight decrease and neutrophil count decrease). **Serious Adverse Events (SAEs):** Reported within 24 hours (29% in Arm A vs. 10% in Arm B). **Data Monitoring Committee (DMC):** Sponsor-managed safety monitoring protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for efficacy endpoints. Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Randomized 42 patients (1:1 ratio) to detect a clinically meaningful difference in pCR between the investigational and active comparator arms. **Handling of Missing Data:** Censoring applied for time-to-event endpoints (EFS, DFS, OS) as per standard Kaplan-Meier methodology.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. Approved by the Medical Ethics Committee of Beijing Cancer Hospital. **Monitoring Plan:** Routine site visits and remote data monitoring to assure protocol compliance. **Audit Trail:** Standard electronic Case Report Form (eCRF) query management and data locking protocols.

11. Study Identifiers & Governance

Primary ID: ML42058 **Registry Number:** NCT04661150 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Sponsor/Lead Organization:** F. Hoffmann-La Roche Ltd (Roche) **Study Title:** A Study of Atezolizumab and Trastuzumab in Combination With Capecitabine and Oxaliplatin in Patients With HER2 Positive Locally Advanced Resectable Gastric Cancer of Adenocarcinoma of Gastroesophageal Junction **Official Regulatory Title:** A Phase II, Randomized Study of Atezolizumab (Anti-PD-L1 Antibody) and Trastuzumab in Combination With Capecitabine and Oxaliplatin (Xelox) in Patients With HER2 Positive Locally Advanced Resectable Gastric Cancer of Adenocarcinoma of Gastroesophageal Junction (GEJ)

12. Clinical Framework (Gastric Cancer Specific)

Disease Areas / Primary Condition: Gastric Cancer, Gastroesophageal Junction Adenocarcinoma **Diagnosis Threshold:** HER2-positive (IHC 3+ or IHC 2+ with ISH $\geq$ 2.0) **Medical Methodology:** Perioperative (Neoadjuvant + Adjuvant) Chemoimmunotherapy **Optimization Target:** Pathological Complete Response (pCR) and R0 Resection

13. Study Procedures & Methodology

Management Pathway: 3 Neoadjuvant Cycles $\rightarrow$ Surgery $\rightarrow$ 5 Adjuvant Cycles **Education Protocol (HCP):** Protocol-specific training for immunotherapy and chemotherapy safety management **Education Protocol (Patient):** Informed consent, surgical prep, and treatment compliance education **Quality Audit Mechanism:** Centralized pathology review for HER2 validation and pCR/MPR assessment

14. Observation & Follow-Up Schedule

Total Duration: Median 15.7 months (Ongoing survival follow-up)
Onsite Visit Cadence: Day 1 of every 21-day cycle (Cycles 1-8)
Remote Monitoring Frequency: Standard inter-cycle outreach
Checklist Review Interval: Post-surgical pathology review and End-of-Treatment evaluation

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				